Project Title

Banking & Loan Risk Analytics System

Domain: Banking & Finance

Technology Stack

- **Excel** – Initial data cleaning and validation
- **SQL** – Database management and advanced analysis
- **Python** – EDA, statistical analysis and ML
- **Power BI** – Interactive dashboards
- **Power Query** – Data transformation

Main Modules

1. Customer Analytics

- Customer demographics
- Income analysis
- Employment status
- Credit history
- Existing loans
- Debt-to-Income Ratio
- Customer risk segmentation

2. Loan Analytics

- Loan applications
- Approved vs. rejected loans
- Loan amount
- Interest rate
- Loan tenure
- EMI
- Outstanding loan amount

3. Credit Risk Analytics

- Default rate
- Delinquency analysis
- High-risk customers
- Credit risk scoring
- NPA analysis
- Early-warning indicators

4. Repayment Analytics

- On-time payments
- Late payments
- Missed payments
- Days Past Due
- Recovery amount
- Collection performance

5. Branch Performance

- Branch-wise loan distribution
- Approval rate
- Default rate
- Outstanding portfolio
- Recovery performance

Advanced SQL Topics

Use:

```
JOINs
CTEs
Subqueries
CASE WHEN
Window Functions
RANK()
DENSE_RANK()
ROW_NUMBER()
LAG()
LEAD()
Running Total
Moving Average
Customer Cohort Analysis
```

Python Analysis

Use:

```
Pandas
NumPy
Matplotlib
Seaborn
Data Cleaning
EDA
Missing Value Treatment
Outlier Detection
Feature Engineering
```

```
Correlation Analysis
Statistical Analysis
```

For an advanced level, add:

Logistic Regression / Random Forest

to build a **Loan Default Prediction Model**.

Power BI Dashboards

Create **5 dashboards**:

1. **Executive Banking Dashboard**
2. **Loan Portfolio Dashboard**
3. **Credit Risk Dashboard**
4. **Customer Risk Dashboard**
5. **Branch & Recovery Dashboard**

Database Tables

Use at least **8 tables**:

```
Customers
Loan_Applications
Loans
Payments
Branches
Employees
Credit_History
Collections
```

Final Project Flow

```
Raw Data
   ↓
Excel / CSV
   ↓
Power Query
   ↓
SQL Database
   ↓
Data Cleaning
   ↓
Advanced SQL Analysis
   ↓
Python EDA
   ↓
```

```
      Machine Learning
         ↓
      Power BI Dashboard
```

This will be a **complete Data Analyst portfolio project**, covering **SQL + Python + Excel + Power BI + Machine Learning**, rather than just a dashboard project.